ABDULSAMAD **ALIYU**

Aberdeen, UK · 07786 626370
abdulsamadsaliyu@gmail.com · linkedin.com/in/abdulsamad-aliyu-shehu-

PROFILE

Builds AI, data and security work that holds up in real use: software backed by automated tests, models measured against baselines, and systems checked for weaknesses before release. Recent work includes a finance platform in daily use, an evaluated text-to-SQL agent and network threat detection. Python, SQL, TypeScript, PyTorch.

TECHNICAL SKILLS

- **Languages:** Python, SQL, TypeScript.
- **AI & ML:** PyTorch, Scikit-learn, LLM APIs, feature engineering, classification and regression, anomaly detection (Isolation Forest), time-series forecasting (SARIMA), evaluation sets, validation with fixed splits, seeds and baselines.
- **Data:** Postgres, Pandas, NumPy, Excel, joining and cleaning data from inconsistent sources, reconciling output against an independent source.
- **Security:** authentication and access-control review, webhook verification, threat detection, encryption (AES-GCM, PBKDF2), Splunk, Scapy.
- **Engineering:** automated testing (253 tests on one application), REST APIs, Git, Docker, Linux, AWS, Azure, Supabase, React, Shopify, SEO.

PROJECTS

2025 – PRESENT

LIVE FINANCE PLATFORM FOR A SELF-EMPLOYED CLIENT, PYTHON, SQL, POSTGRES, TYPESCRIPT, SUPABASE · IN DAILY USE

- Found and fixed six security weaknesses in a live finance platform, including a login with no attempt limit, nine routes reachable without signing in and a webhook that would accept a forged message.
- Caught a tax figure that was wrong while every automated test passed, by reconciling the output against published HMRC guidance, then corrected the logic.
- Shipped four AI features in small releases, then tracked use: two are used constantly and one is barely touched, which reset the roadmap.
- Combined three data sources into one structure: a direct bank feed, photographed receipts read with OCR, and forwarded invoice emails.

2025

NATURAL-LANGUAGE-TO-SQL AGENT WITH ITS OWN EVALUATION, PYTHON, SQL, LLM APIS, SIX RELATIONAL DATABASES, 60-QUESTION GOLD-LABELLED SET

- Scored the agent on a 60-question gold-labelled set by category, which showed which question types it got quietly wrong.
- Built it to read a database schema at runtime, write SQL, and read the error and retry when a query fails.

CUSTOMER LIFETIME VALUE AND CHURN FOR A UK ONLINE RETAILER, PYTHON, PANDAS, SCIPY, SCIKIT-LEARN · 1,067,371 TRANSACTION LINES (UCI ONLINE RETAIL II)

- Predicted churn at 0.73 ROC-AUC with gradient boosting, against 0.68 for a simple 'no order in 90 days' rule, using a time-based backtest with no leakage.
- Wrote the BG/NBD and Gamma-Gamma lifetime-value models from the equations; they ranked 6-month customer value better than gradient boosting (Spearman 0.61 vs 0.50).
- Turned the scores into a retention plan: 2,195 of 4,261 active customers to contact, at an expected £2.27 back per £1 under stated margin and save-rate assumptions.
- Segmented 5,852 customers with RFM, checked the segments against k-means, and mined 512 cross-sell rules from 33,361 baskets.

2024 – 2025

PREDICTIVE MODELLING ON CUSTOMER DATA, PYTHON, SCIKIT-LEARN, 50,000-RECORD DATASETS

- Reached 88% precision on churn across 50,000 customer records; the gain came from multivariate feature engineering rather than a larger model.

- Improved precision by 28% over a baseline phishing classifier on 50,000+ emails, through feature engineering and decision-threshold tuning.

2025 – PRESENT
RECOMMENDATION AND RANKING ENGINE, TYPESCRIPT, INDEXEDDB, 24,063-ITEM CATALOGUE, 253 AUTOMATED TESTS

- Cut search time per keystroke from 7.3ms to 1.6ms by writing a typo-tolerant, accent-insensitive ranker.
- Built a scoring engine that computes a personal match for all 24,063 catalogue items on the user's device.

2024 – 2025
ANOMALY DETECTION AND FORECASTING ON UNLABELLED EVENT DATA, PYTHON, SCIKIT-LEARN (ISOLATION FOREST), SARIMA, 50,000 EVENTS IN SIX FORMATS

- Flagged anomalies across 50,000 unlabelled events arriving in six inconsistent formats, then added severity scoring so they were triaged by risk.

2025 – 2026
CONTROLLED MODEL COMPARISONS, PYTORCH, GYMNASIUM, FIXED 70/15/15 SPLITS AND SEEDS, FULL HYPERPARAMETER SWEEPS

- Showed the smaller EfficientNet-B0 beat EfficientNetV2-S at 96.67% accuracy with a quarter of the parameters, and traced why: a learning-rate mismatch and class imbalance.
- Compared Q-learning and SARSA across 2 environments with a full hyperparameter sweep; the algorithm that converged faster learned riskier policies.

2024 – 2025
NETWORK TRAFFIC ANALYSIS AND SECURE CREDENTIAL STORAGE, PYTHON, SCAPY, AES-GCM, PBKDF2

- Built a network traffic analyser that supports both stateful and stateless threat detection, and a credential store encrypted with AES-GCM.

PROFESSIONAL EXPERIENCE

MAY 2026 – SEP 2026
SERVICE ASSISTANT, SAINSBURY'S, ABERDEEN

- Served customers and covered several store functions in a high-volume store.

FEB 2025 – JUN 2025
WEBSITE DEVELOPMENT & ANALYTICS INTERN, SPÈ CANDLES, REMOTE

- Owned weekly analytics reporting to the owner and turned it into clear actions; engagement rose 25% over the placement.
- Removed duplicate pages that were confusing search engines, rewrote every meta description and restructured the URLs on the Shopify store.
- Recommended Shopify apps and wrote AI prompts to speed up content creation, and documented every change for the owner.

NOV 2021 – JUL 2022
SALES ASSISTANT, CEX, ABERDEEN

- Held 99% accuracy on grading and pricing across 500+ stock items, and redesigned a slow pricing workflow, cutting average service time by 15%.

EDUCATION

SEP 2024 – JUN 2026
MSC ARTIFICIAL INTELLIGENCE, UNIVERSITY OF STIRLING

SEP 2018 – JUL 2022
BSC (HONS) CYBERSECURITY, THE ROBERT GORDON UNIVERSITY, ABERDEEN

CERTIFICATIONS

- **AWS Certified AI Practitioner:** February 2026.
- **Microsoft Azure AI Fundamentals (AI-900):** May 2026.
- **ISC2 Certified in Cybersecurity:** April 2025.